

Panelists



Ashwin Shankar, CEO,
BatteryPool



Naresh Neelkantan, Senior
Architect, Sasken Technologies



Savio Monteiro, Senior Vice
President, Energy, Oil & Gas and
Utilities, Praxis Global Alliance



Shreyas Shibulal, Founder
and Director, Micelio Mobility

structure challenge than what can be perceived from the urban context with 2- and 3-wheelers.

Ashwin believes that setting up battery swapping stations at convenient places would boost the penetration of swappable batteries. The swappable stations could be set up at warehouses, where the drivers of commercial EVs keep coming back.

Battery recycling

Batteries consist of toxic heavy metals, which, when discarded carelessly, can pollute the air and soil. With increase in the number of batteries, a significant problem of battery waste disposal and recycling arises, which can have serious environmental implications.

With popularity of EVs growing exponentially, so would the pile of used Li-ion batteries in the landfills. If present trends for handling spent batteries hold, most of the batteries will end up in landfills, even though these can be recycled.

The batteries consist of valuable metals and other materials that can be recovered from the spent batteries. They can be processed and reused. However, very little recycling goes on today. But with the sustainable energy changes happening around the world, the battery recycling sector is also growing by leaps and bounds.

Shreyas says that degraded batteries are a burden to industries and battery recycling is going to be a tremendous opportunity in the next couple of years by when the first generation of batteries will start arriving for recycling. He thinks, battery recycling

is one of the spaces which would become highly regulated. "There is going to be a sufficient amount of pressure by regulators on the private sectors to pick up their trash and recycle it accordingly." Therefore, regulations would be a driver of creating these ecosystems down the road.

Alternative to lithium-ion

Li-ion, though a standard battery technology today, also has some disadvantages in terms of safety and degradation. Researchers and engineers across the globe are trying to come up with new chemistries that can overcome the challenges of Li-ion batteries and provide performance benefits as well.

According to Shreyas, hybrid ultracapacitor batteries are probably going to emerge in the next couple of years. Among some slightly long-term technologies, solid-state batteries will emerge due to their high energy density and relatively simple thermal management. He also talks about metal-air batteries with aluminium. "India has a substantial amount of deposits of bauxite, from which aluminium is made. This presents itself as an opportunity if metal-air batteries become widespread. However, this technology is in its early stages of development and the energy density and output power of those batteries is quite limited now."

Ashwin says, these advancing technologies must not only be benchmarked against Li-ion batteries in laboratories, but the important steps include getting these technologies from laboratories to the field environment, in order to mass produce

and make them affordable for mass usage. He adds that Li-ion batteries have been with us for the last 40 years now and are today used in almost all energy-storage situations. According to him, Li-ion batteries are here to stay for the foreseeable future.

Battery manufacture in India

The government's target to have an all-electric fleet by 2047 has given a major push to the automotive industries to manufacture batteries in India. Currently, India lacks the battery manufacturing ecosystem and, as a result, the import of batteries is leading to increased drain on foreign exchange. So, this is a wonderful time to establish a battery manufacturing ecosystem so as to cater to the EVs of the future.

Shreyas suggests that the biggest thing would really be securing the raw materials required for the manufacturing process. High degree of indigenous effort is required at this stage for long-term benefits. If we are able to diversify the supply chain, and not be dependent on any particular country, that would be strategically a good move to sustain the battery ecosystem in the country.

As per Ashwin, the only way to make the economics of battery technology work is when there are economies of scale. Thus, creating policies that really spur investment in battery manufacture on a large scale is an essential step. Savio says, it requires convincing private sector companies to take lead in setting up the battery ecosystem, while the government creates policies and incentives to support the ecosystem. **EFY**